CLINICAL DATA ANALYTICS PORTFOLIO

Healthcare Capstone: OCD Patient Cohort Analysis

Demographic trends, symptom severity metrics, and diagnostic timelines evaluated via MySQL advanced window framing and analytical queries.

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MySQL Engine 8.0

EXECUTIVE SUMMARY

Clinical Study Architecture & Objectives

Clinical Diagnostics

Auditing Y-BOCS obsession and compulsion metrics across diverse demographics.

SQL Engineering

Executing 15 business queries leveraging CTEs, window functions, and data hygiene scripts.

Outcome Insights

Delivering actionable trends for hospital management and psychiatric research teams.

DATA MODEL

Database Schema & Table Attributes

Primary Entities (`ocd_patient`)

- **Patient ID:** Unique identifier string
- **Demographics:** Age, Gender, Ethnicity, Marital Status
- **Clinical Scores:** Obsession Score & Compulsion Score
- **Timelines:** Diagnosis Date & Symptom Duration

Data Hygiene Protocols

Audited NULL values across key metric fields prior to query execution.

Standardized string formats and date structures for temporal evaluation.

COHORT VOLUME

Total Patient Cohort Audited

1,500+

Clinical OCD patient records evaluated across multi-year diagnosis timelines

QUERY ANALYSIS 01

Demographic Distribution & Score Variance

Female Patients

Balanced obsession scores with high reporting frequency in early diagnostics.

Male Patients

Elevated compulsion metrics observed in adolescent and young adult brackets.

SECTION 02

Advanced SQL Window Functions & Cumulative Metrics

Exploring running totals and ranking queries applied to clinical datasets.

QUERY ANALYSIS 02

Diagnostic Timeline & Running Totals

Window Frame Logic

`ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` calculates cumulative patient intakes year over year.

```
ORDER BY diag_year ASC
 ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
```

Year	New Patients	Cumulative Total
2018	210	**210**
2019	340	**550**
2020	410	**960**
2021	540	**1,500**

QUERY ANALYSIS 03

Symptom Severity Ranking via DENSE_RANK()

Contiguous Ranking

Ranks clinical severity tiers without skipping sequence numbers on duplicate scores.

High-Severity Cohort

Isolates top-tier obsession index patients for priority therapeutic intervention.

CLINICAL CLASSIFICATION

Prevalent Obsession & Compulsion Categories

Contamination

Washing and sanitizing compulsion patterns.

Symmetry & Order

Arranging and repeating procedural loops.

Harm Thoughts

Checking and verification compulsions.

REPOSITORY PACKAGING

GitHub Modular File Architecture

- **01_schema_setup.sql:** DDL table frames
- **02_data_cleaning.sql:** NULL & hygiene audits
- **03_clinical_queries.sql:** 15 analytical solutions
- **README.md:** Executive portfolio findings

Recruiter Value

Demonstrates production-ready query modularity, clean documentation standards, and structured clinical problem-solving skills.

KEY TAKEAWAYS

Healthcare Strategy & Future Roadmap

Targeted Therapy

Prioritize ERP therapy for high-ranking Y-BOCS score brackets.

Early Intake Protocols

Optimize diagnostic timelines based on cumulative intake metrics.

Thank You & Questions

Healthcare Capstone Project | OCD Clinical Data Analytics

github.com/ahedkhan24-source